CLERK VRAJ NIRAJBHAI

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Professional Summary

Python Developer with professional experience building backend services, reusable Python utilities, REST APIs, and cloud-based applications. Hands-on experience with FastAPI, PostgreSQL, OpenSearch, AWS, and Docker, including enterprise application development, search integration, and scalable backend architecture. Additional experience in deep learning and computer vision using U-Net and ResNet34.

Technical Skills

Languages: Python, JavaScript
Python Backend: FastAPI, RESTful APIs
Databases & Search: PostgreSQL, MySQL, MongoDB, OpenSearch, Amazon DynamoDB
Cloud & DevOps: AWS (Lambda, S3, DynamoDB, API Gateway, EC2, ECR), Docker, Terraform, CI/CD

Experience

Techlusion Dec 2024 – Present
Software Developer Intern → Junior Software Developer *Ahmedabad, India*

- Developed an internal Python package that encapsulates OpenSearch queries into reusable utilities, simplifying search integration and promoting code reuse across development teams.
- Designed and implemented a scalable Master-Child Material Architecture for an enterprise material management platform, migrating legacy data models to hierarchical attributes with audit tracking through DynamoDB Streams and OpenSearch indexing.
- Built and maintained AWS infrastructure using Lambda, DynamoDB, S3, API Gateway, EC2, and ECR to support backend services and application deployments.
- Implemented Auth0-based authentication and role-based access control (RBAC) across multiple applications, establishing reusable authorization patterns.
- Developed an equipment visualization and data synchronization module with interactive charts and high-throughput Excel imports, reducing manual data-entry effort.

Projects

Competitive Pickleball Matchmaking & Scoring Platform June 2025 – Present

- Developed a real-time matchmaking and ranking engine with custom player scoring metrics, team rating calculations, and geolocation-based leaderboard generation.
- Conducted load testing using Locust with 800+ concurrent users, identified database bottlenecks, and offloaded ranking updates to background queues to reduce end-of-match latency.

Lixor: Bottle & Liquid Level Detection System Apr 2025 – May 2025

- Developed a semantic segmentation model using U-Net with a ResNet34 backbone to detect bottles and identify liquid levels in images.
- Implemented image annotation and mask preprocessing pipelines using LabelMe and trained the model using a combined Cross-Entropy and Dice Loss function.

Tiffin Booking System (Tiffin Buddy) Dec 2023 – Mar 2024

- Developed a MERN stack web platform to automate meal subscriptions, scheduling, and delivery workflows.
- Designed relational database schemas in MySQL to manage customer profiles, order calendars, and transaction logs.

Education

Dharmsinh Desai University Gujarat, India
Bachelor of Technology in Information Technology (CPI: 7.7 / 10.0) *2021 – 2025*

Certifications

AWS Academy Machine Learning Foundations | *AWS Academy* 2024
AWS Academy Cloud Foundations | *AWS Academy* 2024